

Solution

National University of Computer and Emerging Sciences, Lahore Campus



Course: Digital Logic Design
Program: BS(Computer Science)
Due Date: 10th March, 2026
Section: BSE-6B

Course Code: AI-2002
Semester: Spring 2026
Weight
Roll No.

Quiz # 3

Section:

You are designing a genetic algorithm for a *minimization problem* that selects individuals for the next generation using Rank based Selection. The probability of selecting an individual is based on its fitness value. Use the following

$$P_i = \frac{2 \times (\text{Rank of Individual})}{N \times (N + 1)}$$

where N is the total number of individuals.

A) Given the following population with their respective fitness values, perform Rank based Selection.

Chromosome #	Fitness	Rank	Probabilities	Range
A	41	4	0.27	0.0 — 0.27
B	13	5	0.33	0.28 — 0.60
C	95	1	0.07	0.61 — 0.67
D	53	3	0.20	0.68 — 0.87
E	89	2	0.13	0.88 — 1.00

You can round the answer to 2 decimal places → B will be selected.

* Ranking is done as $\begin{cases} \text{min fitness} \rightarrow \text{highest rank} \\ \text{max fitness} \rightarrow \text{lower rank} \end{cases}$

bez of minimization problem